Drawing Analysis Results

AI Analysis Summary

Plan View: R3.0 (page 3)

Scale: 1:70

roof_drains: 2

Detail refs: Detail 2/R4.0, Detail 3/R4.0, Detail 4/R4.0, Detail 5/R4.0, Detail 2/R3.0

Details: R2.0 (page 2)

> **DEMOLITION PLAN** [opening_cover] - measured in: sqft
>
> - 1. EXISTING PERIMETER GUARDRAILS → Metal_Panel (EXISTING PERIMETER GUARDRAILS TO REMAIN. TAKE CARE TO PRESERVE EXISTING GUARDRAIL CONDITION. CONTRACTOR RESPONSIBLE FOR ANY DAMAGE TO EXISTING GUARDRAIL DURING CONSTRUCTION)
> - 2. EXISTING SEATING, INCLUDING CONCRETE SUPPORT BLOCKS → Wood_Blocking_Lumber (REMOVE AND DISPOSE OF EXISTING SEATING, INCLUDING CONCRETE SUPPORT BLOCKS)
> - 3. ENTIRE CONCRETE TOPPING → Concrete Pavers (REMOVE AND DISPOSE OF ENTIRE CONCRETE TOPPING, TYP.)
> - 4. EXISTING ASPHALT ROOFING MEMBRANE → SBS_Membrane (REMOVE EXISTING ASPHALT ROOFING MEMBRANE FROM WITHIN PLANTERS)
> - 5. EXISTING PLANTER WALLS AND WATERPROOFING → Parapet_Flashing_Detail (REMOVE AND DISPOSE OF EXISTING PLANTER WALLS AND WATERPROOFING DOWN TO STRUCTURAL SLAB. INCLUDING PARAPET TRANSITION FLASHINGS)

> **PLANTER WALL DEMO DETAIL** [opening_cover] - measured in: linear_ft
>
> - 1. STRUCTURAL CONCRETE SLAB → Plywood_Sheathing (EXISTING STRUCTURAL SLAB TO REMAIN)
> - 2. RUBBERIZED ASPHALTIC WATERPROOF MEMBRANE → Base_Membrane (REMOVE AND DISPOSE OF TYPICAL EXISTING ROOF ASSEMBLY - RUBBERIZED ASPHALTIC WATERPROOF MEMBRANE)
> - 3. POLYETHYLENE SEPARATION SHEET → Polyethylene Film (REMOVE AND DISPOSE OF TYPICAL EXISTING ROOF ASSEMBLY - POLYETHYLENE SEPARATION SHEET)
> - 4. ±38mm XPS INSULATION → XPS_Insulation (REMOVE AND DISPOSE OF TYPICAL EXISTING ROOF ASSEMBLY - ±38mm XPS INSULATION)
> - 5. ±100mm CONCRETE TOPPING R/W W.M.M. → Concrete Pavers (REMOVE AND DISPOSE OF TYPICAL EXISTING ROOF ASSEMBLY - ±100mm CONCRETE TOPPING R/W W.M.M.)
> - 6. EXISTING FLASHINGS → Flashing_General (REMOVE AND DISPOSE OF EXISTING FLASHINGS)
> - 7. EXISTING PLANTER WALLS → Concrete Pavers (REMOVE AND DISPOSE OF EXISTING PLANTER WALLS)
> - 8. EXISTING PLANTER WATERPROOFING → Base_Membrane (REMOVE AND DISPOSE OF EXISTING PLANTER WATERPROOFING)

- 9. EXISTING PLANTER WALL REINFORCING → Fasteners (CUT OFF EXISTING PLANTER WALL REINFORCING FLUSH TO SLAB SURFACE)

Details: R4.0 (page 4)

TYPICAL WALL UPTURN DETAIL [parapet] - measured in: linear_ft

- 1. EXISTING GLAZING TO REMAIN → Sealant_General (existing to remain)
- 2. REMOVE AND REPLACE EXISTING SEALANT AT GLAZING TO PRECAST TRANSITION → Sealant_General (remove and replace existing sealant)
- 3. PROVIDE 12mm DEEP SAWCUT REGLET FOR FLASHING TERMINATION. INCLUDE BEND IN FLASHING TO LOCK INTO REGLET. FILL WITH SEALANT. → Sealant_General (12mm deep sawcut reglet, fill with sealant)
- 4. PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER TERMINATION BAR. FASTEN AT BASE WITH 6.4mm ø PIN BOLTS EMBEDDED 38mm @ MAX. 400mm O/C, FILL TOP LIP WITH SEALANT. → Flashing_General (24ga pre-finished steel flashing)
- 5. SEAL TOP EDGE WITH COMPATIBLE SEALANT → Sealant_General (compatible sealant)
- 6. TERMINATE MEMBRANE AT UPTURN WITH NEW 25mm CONT. CLAMPING BAR. FASTEN WITH 38mm HEX HEAD SCREWS @ 400mm O/C → Flashing_General (25mm continuous clamping bar)
- 7. 38mm HEX HEAD SCREWS → Screws (38mm hex head screws @ 400mm O/C)
- 8. PROVIDE 26GA GALVANIZED STEEL BREAKSHAPE TO FACILITATE MEMBRANE TRANSITION. FASTEN INTO PRE-CAST WITH 6.4mm ø PIN BOLTS @ 300mm O/C. → Flashing_General (26ga galvanized steel brakeshape)

TYPICAL PARAPET UPTURN DETAIL [parapet] - measured in: linear_ft

- 1. EXISTING PRECAST CONCRETE PARAPET PANEL TO REMAIN → Sealant_General (existing to remain)
- 2. EXISTING GUARDRAIL TO REMAIN → Sealant_General (existing to remain)
- 3. NEW SELF ADHERED AIR/VAPOUR BARRIER MEMBRANE AT SLAB TO PRECAST TRANSITION → Vapour_Barrier_Membrane (self adhered air/vapour barrier membrane)
- 4. FILL VOID WITH MINERAL WOOL INSULATION → Batt_Insulation (mineral wool insulation)
- 5. PROVIDE NEW 18GA STEEL STUD CURB WALL → Wood_Blocking_Lumber (18ga steel stud curb wall)
- 6. COVER WITH 13mm PLYWOOD SHEATHING SCREWED AT TOP AND BOTTOM @ 300mm O/C. → Plywood_Sheathing (13mm plywood sheathing)
- 7. PROVIDE STRIP OF NEW SELF-ADHERED AIR/VAPOUR BARRIER MEMBRANE AT INSIDE FACE OF CURB → Vapour_Barrier_Membrane (self-adhered air/vapour barrier membrane strip)
- 8. SELF-ADHERED SANDED STRIPPING MEMBRANE APPLIED HORIZONTALLY TO MINIMIZE JOINTS. LEAVE 25mm LOOP AND FILL WITH BACK ROD. → Base_Membrane_Peel_Stick (self-adhered sanded stripping membrane, 25mm loop)
- 9. COVER WITH SELF-ADHERED CAP SHEET MEMBRANE. PRIME BASE SHEET PRIOR TO APPLICATION. → Cap_Membrane (self-adhered cap sheet membrane)
- 10. TERMINATE MEMBRANE AT UPTURN WITH NEW 25mm CONT. CLAMPING BAR. FASTEN WITH 38mm HEX HEAD SCREWS @ 400mm O/C → Flashing_General (25mm continuous clamping bar)

- 11. PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER CURB → Flashing_General (24ga pre-finished steel flashing)
- 12. PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER TERMINATION BAR. → Flashing_General (24ga pre-finished steel flashing)

EXISTING ROOF DRAIN WATERPROOFING [drain] - measured in: each

- 1. REMOVE AND REPLACE EXISTING ROOF DRAIN PER SPECIFICATION SECTION 22 01 10 → Roof_Drain (remove and replace existing roof drain)
- 2. CHIP OUT EXISTING CONCRETE SLAB TO FACILITATE DRAIN BODY REMOVAL AND REPLACEMENT. POUR NEW CONCRETE TO SUPPORT DRAIN BODY → Sealant_General (chip out existing concrete slab... pour new concrete)
- 3. PROVIDE NEW DRAIN PIPE AS REQUIRED TO TIE INTO EXISTING ROOF DRAINAGE SYSTEM. → Gutter_Downpipe (new drain pipe)
- 4. PROVIDE MINERAL WOOL INSULATION AND 3HR FIRE RATED SEALANT JOINT BETWEEN NEW DRAIN BODY AND CONCRETE SLAB → Batt_Insulation (mineral wool insulation)
- 5. 3HR FIRE RATED SEALANT JOINT → Sealant_General (fire rated sealant joint)
- 6. WOOD BLOCKING AT DRAIN PERIMETER FOR FLANGE SUPPORT → Wood_Blocking_Lumber (wood blocking)
- 7. NEW UNDERDECK CLAMPING RING → Clips (underdeck clamping ring)
- 8. BACK WRAP VAPOUR BARRIER → Vapour_Barrier_Membrane (vapour barrier)
- 9. SET DRAIN INTO PERIMETER SEALANT COMPATIBLE WITH ROOF MEMBRANE → Sealant_General (perimeter sealant)
- 10. FLEECE REINFORCED PMMA MEMBRANE WITHIN SUMP AND 200mm PAST ON ALL SIDES TO COVER CAP SHEET LAPS → Fleece_Reinforcement_Fabric (fleece reinforced PMMA membrane, 200mm past sump)
- 11. SECURE MEMBRANE WITH CLAMPING RING → Clips (clamping ring)
- 12. DRAIN DOME → Roof_Drain (drain dome)

TYPICAL PRECAST CONCRETE PANEL JOINT SEALANT REPLACEMENT [expansion_joint] - measured in: linear_ft

- 1. COMPLETELY REMOVE EXISTING SEALANT FROM JOINT SURFACES → Sealant_General (completely remove existing sealant)
- 2. CLOSED CELL BACKING ROD → Sealant_General (closed cell backing rod)
- 3. NEW SEALANT (WHERE 'A' IS LESS THAN 12mm, GRIND OUT JOINTS FULL DEPTH) → Sealant_General (new sealant, 12mm joint width)

DOOR THRESHOLD DETAIL [opening_cover] - measured in: linear_ft

- 1. REMOVE AND REPLACE EXISTING DOOR THRESHOLD AS REQUIRED → Sealant_General (remove and replace existing door threshold as required)
- 2. NEW COMPRESSIBLE NEOPRENE GASKET BELOW DOOR THRESHOLD, CONTINUOUS ALONG THE FULL DOOR LENGTH → Sealant_General (compressible neoprene gasket)
- 3. PROVIDE FLEECE REINFORCED LIQUID APPLIED MEMBRANE BELOW DOOR THRESHOLD → Fleece_Reinforcement_Fabric (fleece reinforced liquid applied membrane)

- 4. ADDITIONAL LAYER OF CAP SHEET MEMBRANE AT DOOR THRESHOLD, SEE PLAN → Cap_Membrane (additional layer of cap sheet membrane)

+15 WALL UPTURN DETAIL [parapet] - measured in: linear_ft

- 1. EXISTING ROOF ASSEMBLY BELOW +15 BRIDGE TO REMAIN → Sealant_General (existing roof assembly below +15 bridge to remain)
- 2. REMOVE 50mm OF EXISTING INSULATION BELOW CONCRETE TOPPING AND PACK VOID W/ GRANULAR BENTONITE → Sealant_General (remove 50mm of existing insulation... pack void w/ granular bentonite)
- 3. SAWCUT EXISTING CONCRETE TOPPING AT TRANSITION BELOW +15 BRIDGE TO PROVIDE SMOOTH EDGE → Sealant_General (sawcut existing concrete topping)
- 4. NEW 13mm GLASS FACED GYPSUM SHEATHING. FASTEN INTO EXISTING CONCRETE TOPPING TO REMAIN BELOW +15 BRIDGE AT 400mm O/C. → Gypsum_Fiber_Coverboard (13mm glass faced gypsum sheathing)
- 5. LAP NEW VAPOUR BARRIER MEMBRANE MIN. 200mm ONTO EXISTING MEMBRANE BELOW +15 BRIDGE. → Vapour_Barrier_Membrane (lap new vapour barrier membrane min. 200mm)
- 6. CONTINUE VAPOUR BARRIER MEMBRANE MIN. 50mm ABOVE INSULATION TO FACILITATE TIE-IN WITH BASE SHEET MEMBRANE → Vapour_Barrier_Membrane (continue vapour barrier membrane min. 50mm above insulation)
- 7. TEMPORARILY REMOVE AND REINSTATE EXISTING METAL CLADDING TO FACILITATE ROOF MEMBRANE UPTURN → Metal_Panel (temporarily remove and reinstate existing metal cladding)

Specification PDF Analysis (Deterministic Regex Engine)

20 unique products found across **7** categories — **10** mapped to pricing keys.
Project: Replacement

▶ **Membranes (5)**

▶ **Sealants & Adhesives (7)**

▶ **Drainage (3)**

▶ **Wood & Sheathing (1)**

▶ **Miscellaneous (1)**

▶ **Insulation (1)**

▶ **Metal Flashings & Accessories (2)**

Spec-Driven Pricing Estimate (Drawing spatial data + Specification materials — deterministic)

LINE ITEMS RESOLVED	RESOLUTION FAILURES	TOTAL MATERIAL COST
3	**0**	**$1,062.54**

Resolved Line Items

DETAIL	TYPE	CONFIRMED MATERIAL	QTY	UNIT PRICE	LINE COST	SOURCE
TYPICAL WALL UPTURN DETAIL R4.0	parapet	Sealant (General) `Sealant_General` spec pg 2, 86, 88, 89, 90, 91, 92, 93, 94, 95	8 tube	$29.86	$238.88	FAL
TYPICAL PARAPET UPTURN DETAIL R4.0	parapet	Vapour Barrier Membrane `Vapour_Barrier_Membrane` spec pg 68	1 roll	$173.66	$173.66	FAL
EXISTING ROOF DRAIN WATERPROOFING R4.0	drain	Roof Drain `Roof_Drain` spec pg 2, 13, 29, 56, 63, 73, 82, 96, 97, 98, 99	2 EA	$325.00	$650.00	FAL
				Total Material Cost	**$1,062.54**	

Detail-Based Quantity Takeoff (AI-Analyzed)

ROOF AREA	PERIMETER	PARAPET	PENETRATIONS
1,534 sqft	**230 LF**	**139 LF x 0.66 ft**	**0**

Field Assembly (Plan View) [field_assembly] (ref: ?)

Measured in: sqft | Base value: 1,534.0 | Detail cost: $23,967.14 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
Asphaltic Primer	7 pail	$282.83	$1,979.81
SBS Base Sheet - Field (Sopraply Base 520)	18 roll	$127.25	$2,290.50
SBS Base Sheet - Wall (Sopraply Base 520)	2 roll	$127.25	$254.50
SBS Cap Sheet - Field (Sopraply Traffic Cap)	21 roll	$105.80	$2,221.80
SBS Cap Sheet - Wall (Sopraply Traffic Cap)	2 roll	$105.80	$211.60
Tapered ISO Insulation (drainage slope)	1688 sqft	$3.10	$5,232.80
XPS Insulation (Sopra-XPS 40 Type 4)	106 sheet (2'x8')	$43.70	$4,632.20
Drainage Board (Sopradrain EcoVent)	6 roll (6'x50')	$410.53	$2,463.18
Filter Fabric	6 roll	$150.00	$900.00
Soprasmart ISO HD 1/2" (factory laminated coverboard)	71 sheet (3'x8')	$53.25	$3,780.75

TYPICAL WALL UPTURN DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 139.0 | Detail cost: $1,518.80 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
REMOVE AND REPLACE EXISTING SEALANT AT GLAZING TO PRECAST TRANSITION	8 tube	$29.86	$238.88
PROVIDE 12mm DEEP SAWCUT REGLET FOR FLASHING TERMINATION. INCLUDE BEND IN FLASHING TO LOCK INTO REGLET. FILL WITH SEALANT.	0 tube	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER TERMINATION BAR. FASTEN AT BASE WITH 6.4mm ø PIN BOLTS EMBEDDED 38mm @ MAX. 400mm O/C, FILL TOP LIP WITH SEALANT.	16 10ft piece	$75.00	$1,200.00
SEAL TOP EDGE WITH COMPATIBLE SEALANT	0 tube	$0.00	-
TERMINATE MEMBRANE AT UPTURN WITH NEW 25mm CONT. CLAMPING BAR. FASTEN WITH 38mm HEX HEAD SCREWS @ 400mm O/C	0 10ft piece	$0.00	-
38mm HEX HEAD SCREWS	2 box	$39.96	$79.92
PROVIDE 26GA GALVANIZED STEEL BREAKSHAPE TO FACILITATE MEMBRANE TRANSITION. FASTEN INTO PRE-CAST WITH 6.4mm ø PIN BOLTS @ 300mm O/C.	0 10ft piece	$0.00	-

TYPICAL PARAPET UPTURN DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 139.0 | Detail cost: $1,282.39 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
NEW SELF ADHERED AIR/VAPOUR BARRIER MEMBRANE AT SLAB TO PRECAST TRANSITION	1 roll	$173.66	$173.66
FILL VOID WITH MINERAL WOOL INSULATION	3 bundle	$83.99	$251.97
PROVIDE NEW 18GA STEEL STUD CURB WALL	20 8ft piece	$16.59	$331.80
COVER WITH 13mm PLYWOOD SHEATHING SCREWED AT TOP AND BOTTOM @ 300mm O/C.	4 4'x8' sheet	$48.74	$194.96
PROVIDE STRIP OF NEW SELF-ADHERED AIR/VAPOUR BARRIER MEMBRANE AT INSIDE FACE OF CURB	0 roll	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
SELF-ADHERED SANDED STRIPPING MEMBRANE APPLIED HORIZONTALLY TO MINIMIZE JOINTS. LEAVE 25mm LOOP AND FILL WITH BACK ROD.	2 roll	$165.00	$330.00
COVER WITH SELF-ADHERED CAP SHEET MEMBRANE. PRIME BASE SHEET PRIOR TO APPLICATION.	0 roll	$0.00	-
TERMINATE MEMBRANE AT UPTURN WITH NEW 25mm CONT. CLAMPING BAR. FASTEN WITH 38mm HEX HEAD SCREWS @ 400mm O/C	0 10ft piece	$0.00	-
PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER CURB	0 10ft piece	$0.00	-
PROVIDE 24GA PRE-FINISHED STEEL FLASHING TO COVER TERMINATION BAR.	0 10ft piece	$0.00	-

EXISTING ROOF DRAIN WATERPROOFING [drain] (ref: R4.0)

Measured in: each | Base value: 2 | Detail cost: $663.60 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
REMOVE AND REPLACE EXISTING ROOF DRAIN PER SPECIFICATION SECTION 22 01 10	2 EA	$325.00	$650.00
CHIP OUT EXISTING CONCRETE SLAB TO FACILITATE DRAIN BODY REMOVAL AND REPLACEMENT. POUR NEW CONCRETE TO SUPPORT DRAIN BODY	0 tube	$0.00	-
PROVIDE NEW DRAIN PIPE AS REQUIRED TO TIE INTO EXISTING ROOF DRAINAGE SYSTEM.	1 EA	$13.60	$13.60
PROVIDE MINERAL WOOL INSULATION AND 3HR FIRE RATED SEALANT JOINT BETWEEN NEW DRAIN BODY AND CONCRETE SLAB	0 bundle	$0.00	-
3HR FIRE RATED SEALANT JOINT	0 tube	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
WOOD BLOCKING AT DRAIN PERIMETER FOR FLANGE SUPPORT	0 8ft piece	$0.00	-
NEW UNDERDECK CLAMPING RING	0 piece	$10.19	-
BACK WRAP VAPOUR BARRIER	0 roll	$0.00	-
SET DRAIN INTO PERIMETER SEALANT COMPATIBLE WITH ROOF MEMBRANE	0 tube	$0.00	-
FLEECE REINFORCED PMMA MEMBRANE WITHIN SUMP AND 200mm PAST ON ALL SIDES TO COVER CAP SHEET LAPS	0 roll	$0.00	-
SECURE MEMBRANE WITH CLAMPING RING	0 piece	$0.00	-
DRAIN DOME	0 EA	$0.00	-

TYPICAL PRECAST CONCRETE PANEL JOINT SEALANT REPLACEMENT [expansion_joint] (ref: R4.0)

Measured in: linear_ft | Base value: 57.5 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
COMPLETELY REMOVE EXISTING SEALANT FROM JOINT SURFACES	0 tube	$0.00	-
CLOSED CELL BACKING ROD	0 tube	$0.00	-
NEW SEALANT (WHERE 'A' IS LESS THAN 12mm, GRIND OUT JOINTS FULL DEPTH)	0 tube	$0.00	-

DOOR THRESHOLD DETAIL [opening_cover] (ref: R4.0)

Measured in: each | Base value: 0 | Detail cost: $0.00 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
REMOVE AND REPLACE EXISTING DOOR THRESHOLD AS REQUIRED	0 tube	$0.00	-
NEW COMPRESSIBLE NEOPRENE GASKET BELOW DOOR THRESHOLD, CONTINUOUS ALONG THE FULL DOOR LENGTH	0 tube	$0.00	-
PROVIDE FLEECE REINFORCED LIQUID APPLIED MEMBRANE BELOW DOOR THRESHOLD	0 roll	$0.00	-
ADDITIONAL LAYER OF CAP SHEET MEMBRANE AT DOOR THRESHOLD, SEE PLAN	0 roll	$0.00	-

+15 WALL UPTURN DETAIL [parapet] (ref: R4.0)

Measured in: linear_ft | Base value: 139.0 | Detail cost: $118.48 qty: fallback

MATERIAL	QTY	UNIT PRICE	LINE COST
EXISTING ROOF ASSEMBLY BELOW +15 BRIDGE TO REMAIN	0 tube	$0.00	-
REMOVE 50mm OF EXISTING INSULATION BELOW CONCRETE TOPPING AND PACK VOID W/ GRANULAR BENTONITE	0 tube	$0.00	-
SAWCUT EXISTING CONCRETE TOPPING AT TRANSITION BELOW +15 BRIDGE TO PROVIDE SMOOTH EDGE	0 tube	$0.00	-
NEW 13mm GLASS FACED GYPSUM SHEATHING. FASTEN INTO EXISTING CONCRETE TOPPING TO REMAIN BELOW +15 BRIDGE AT 400mm O/C.	4 sheet (4'x8')	$29.62	$118.48
LAP NEW VAPOUR BARRIER MEMBRANE MIN. 200mm ONTO EXISTING MEMBRANE BELOW +15 BRIDGE.	0 roll	$0.00	-

MATERIAL	QTY	UNIT PRICE	LINE COST
CONTINUE VAPOUR BARRIER MEMBRANE MIN. 50mm ABOVE INSULATION TO FACILITATE TIE-IN WITH BASE SHEET MEMBRANE	0 roll	$0.00	-

Total Material Cost **$27,550.41**

Standard Estimate (for comparison)

Standard Total Estimate **$54,470.75**